Econ 453 Data Analytics and Modeling: Quantitative Analysis for Economic Strategy

Spring 2026
The University of Arizona
Department of Economics

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Class Day/Time: Mon/Wed 3:30 PM – 4:45 PM
Class Location: McClelland Hall 128

Course Description

This course provides an introduction to core statistical tools and modern machine learning methods commonly used in the analysis of business and economic data. Students will learn canonical estimation techniques, including linear regression, logistic regression, linear discriminant analysis, cross-validation, bootstrap methods, non-parametric and high-dimensional regression, and random forests. Emphasis will be placed on real-world applications using business data and on implementation in `R`. Essential concepts and tools from probability and statistics—such as probability distributions, descriptive statistics, data cleaning and visualization, confidence-interval construction, hypothesis testing, and the principles of maximum likelihood—will be introduced as needed. These topics serve to help students understand the connection between traditional inferential statistics and contemporary machine learning methods.

Expected Learning Outcomes

Upon successful completion of this course, students will be able to clean and summarize data using visualization and descriptive analytics; apply appropriate data-mining techniques and estimation methods for predictive analysis and decision making; conduct basic statistical inference to assess estimation accuracy; program in `R` for all the above objectives; and interpret and communicate estimation results produced by a range of machine learning algorithms.

Prerequisites

(ECON 300 or ECON 361) and (AREC/ECON 339 or BNAD 276). Students should have prior exposure to basic probability, statistics, and algebra. Any additional concepts from calculus or matrix algebra needed for the course will be introduced as they arise. No prior programming experience is required.

Required Textbook

In principle, the lecture slides will be self-contained and no additional resources are required.

However, the structure of this course will closely follow *An Introduction to Statistical Learning: With Applications in R*, 2nd edition, by James, Witten, Hastie, and Tibshirani (2021), Springer. I highly recommend reading the corresponding sections before coming to class. This book is freely available online through the UA Library.

Statistical Software

We will use `R` in this course. `R` is a free, open-source programming language and environment for statistical computing and graphics. There are tons of `R` tutorials online, including the ones that accompany our textbook. Also, keep in mind that almost any R coding question you run into has already been asked and answered somewhere on the internet. So Google or ChatGPT may be your best friend here. And if you're still running into issues, please don't hesitate to reach out to me or the TA.

How to Succeed in This Course?

Although I will deemphasize the more theoretical inferential components and focus primarily on application and implementation, data analysis is inherently mathematical, which means it is **HARD**. Understanding and internalizing these concepts requires repetition and study outside the classroom. Depending on your background, you should expect to spend approximately **5–15 hours** per week to fully absorb each week's material.

Grading Policy

The final course grade will be based on the following components:

- **Assignment 1 (5%)**: Regression Methods.
- **Assignment 2 (5%)**: Classification Methods.
- **Exam 1 (30%)**: In class.
- **Exam 2 (30%)**: In class.
- **Group Project (20%)**: TBA.
- **Class attendance (10%)**: Each student is allowed up to three unexcused absences.

Grading scale: A: 100–90%, B: 89–80%, C: 79–70%, D: 69–60%, and E: below 60%.
Late work will not be accepted unless official accommodations apply.

Tentative Schedule

All readings refer to the required textbook.

Week	Dates	Topic	Reading
1	Jan 14	Overview	ISLR Ch 1
2	Jan 19 (No class)/21	Intro. to `R`	—
3	Jan 26/28	Basic Probability	—
4	Feb 2/4	Basic Probability	—
5	Feb 9/11	Statistical Tools	—
6	Feb 16/18	Statistical Tools	—
7	Feb 23/25	Simple Linear Regression	ISLR Ch 3.1
8	Mar 2/4	Review & Exam 1	—
9	Mar 9/11 (No class)	—	—
10	Mar 16/18	Model Inference	ISLR Ch 3.1
11	Mar 23/25	Matrix Notation & Multiple Regression	ISLR Ch 3.2
12	Mar 30/Apr 1	Non-Parametric & Ridge & Lasso	ISLR Ch 6.1 6.2 7.1
13	Apr 6/8	Logistic Regression & LDA	ISLR Ch 4.3 4.4
14	Apr 13/15	Cross validation & Bootstrap	ISLR Ch 5.1 5.2
15	Apr 20/22	Review & Exam 2	—
16	Apr 27/29	Tree & Random Forrest	ISLR Ch 8.1 8.2
17	May 4/6	Project Presentation	—

University Policies

Code of Academic Integrity

Students are encouraged to share intellectual views and discuss freely the principles and applications of course materials. However, graded work/exercises must be the product of independent effort unless otherwise instructed. Students are expected to adhere to the UA Code of Academic Integrity as described in the UA General Catalog. `https://deanofstudents.arizona.edu/policies/code-academic-integrity`.

Accommodations for Students with Disabilities

At the University of Arizona, we strive to make learning experiences as accessible as possible. If you anticipate or experience barriers based on disability or pregnancy, please contact the Disability Resource Center (520-621-3268, `https://drc.arizona.edu/`) to establish reasonable accommodations.

Honors Contract

Students wishing to contract this course for Honors Credit should email the Professor to discuss the terms of the contract. Information on Honors Contracts can be found at `https://www.honors.arizona.edu/honors-contracts`.

Safety on Campus and in the Classroom

For a list of emergency procedures for all types of incidents, please visit the website of the Critical Incident Response Team (CIRT): `https://cirt.arizona.edu/about-cirt/overview`.

Threatening Behavior Policy

The UA Threatening Behavior by Students Policy prohibits threats of physical harm to any member of the University community, including to oneself. `http://policy.arizona.edu/education-and-student-affairs/threatening-behavior-students`.

Nondiscrimination and Anti-harassment Policy

The University of Arizona is committed to creating and maintaining an environment free of discrimination. In support of this commitment, the University prohibits discrimination, including harassment and retaliation, based on a protected classification, including race, color, religion, sex, national origin, age, disability, veteran status, sexual orientation, gender identity, or genetic information. `http://policy.arizona.edu/human-resources/nondiscrimination-and-anti-harassment-policy`.

Absence and Class Participation Policies

Absences for any sincerely held religious belief, observance, or practice will be accommodated where reasonable. Refer to the Accommodation Policy: https://policy.arizona.edu/human-resources/religious-accommodation-policy. Absences pre-approved by the University Dean of Students (or dean's designee) will be honored.

University Policies

All university policies related to a syllabus are available at: https://catalog.arizona.edu/syllabus-policies.

Additional Resources for Students

UA Academic policies and procedures: http://catalog.arizona.edu/policies. Student Assistance and Advocacy: https://deanofstudents.arizona.edu/support/student-assistance.

Subject to Change Notice

Information contained in the course syllabus, other than the grade and absence policies, may be subject to change with reasonable advance notice, as deemed appropriate by the instructor of this course.